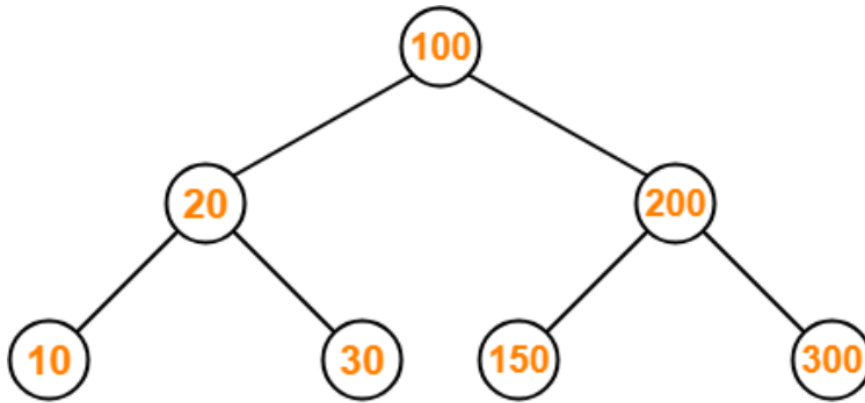


CO3- AT3- Tree Traversal Challenges

Write the traversal sequences for following trees

Question No :1



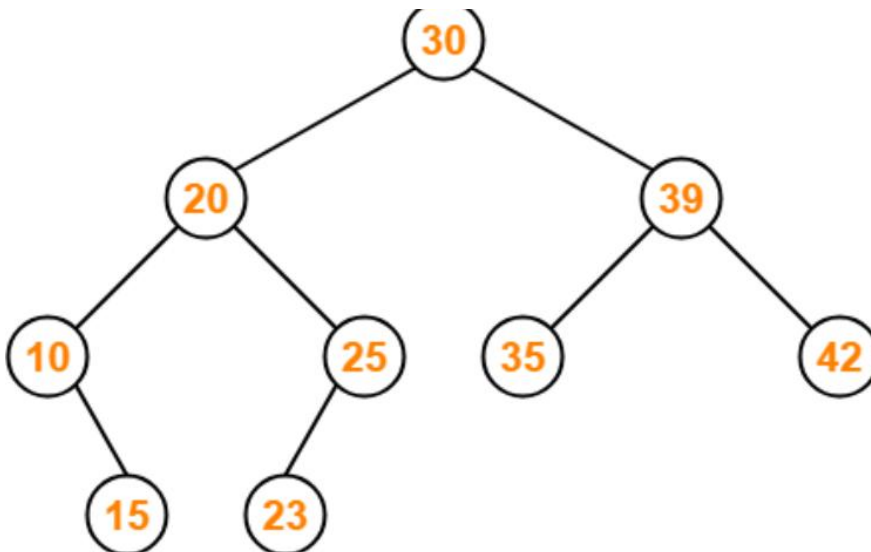
Ans:

Inorder: 10, 20, 30, 100, 150, 200, 300

Preorder: 100, 20, 10, 30, 200, 150, 300

Postorder: 10, 30, 20, 150, 300, 200, 100

Question No : 2



Ans:

Inorder: 2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20

Preorder: 15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20

Postorder: 2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15

Question No : 3

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

Ans:

Numbers inserted: 7, 5, 1, 8, 3, 6, 0, 9, 4, 2

Inorder: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Preorder: 7, 5, 1, 0, 3, 2, 4, 6, 8, 9

Postorder: 0, 2, 4, 3, 1, 6, 5, 9, 8, 7

Question No : 5

```
      1
     /\
    2 3
   /\ \
  4 5 6
   /
  7
```

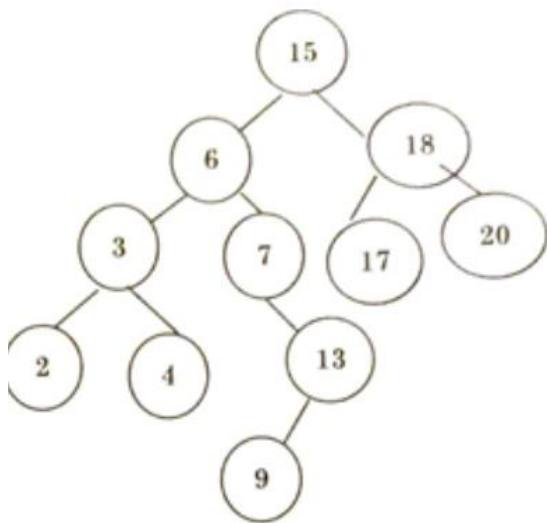
Ans:

Inorder: 4, 2, 7, 5, 1, 3, 6

Preorder: 1, 2, 4, 5, 7, 3, 6

Postorder: 4, 7, 5, 2, 6, 3, 1

Question No 6



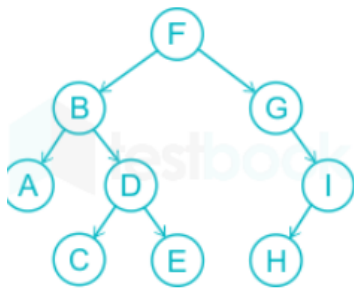
Ans:

- Inorder: **D B E F A G H C**
- Preorder: **A B D E F C G H**

Postorder: D, F, E, B, H, G, C, A

Question No 7

What is the sequence of nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G

2. A, C, D, E, B, F, G, H, I

3. A, B, C, D, E, F, G, H, I

4. I, H, G, F, E, D, C, B, A

Ans:

Preorder: A, B, E, F, I, J, K, C, G, H, D

Postorder: E, I, J, K, F, B, G, H, C, D, A

Question No 8

Consider the following Inorder and Preorder traversal of a tree

Inorder – D B E F A G H C

Preorder - A B D E F C G H

What is the Postorder Traversal of the same tree

1. D F E B H C G A

2. D F E B G H C A

3. D F E B H G C A

4. D F E H B G C A

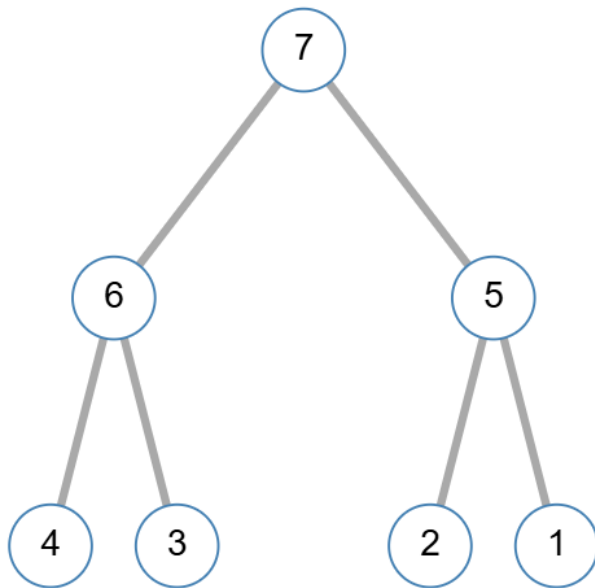
Ans:

Inorder: 10, 15, 20, 23, 25, 30, 35, 39, 42

Preorder: 30, 20, 10, 15, 25, 23, 39, 35, 42

Postorder: 15, 10, 23, 25, 20, 35, 42, 39, 30

Question No 9 : Find Inorder, Preorder and Postorder



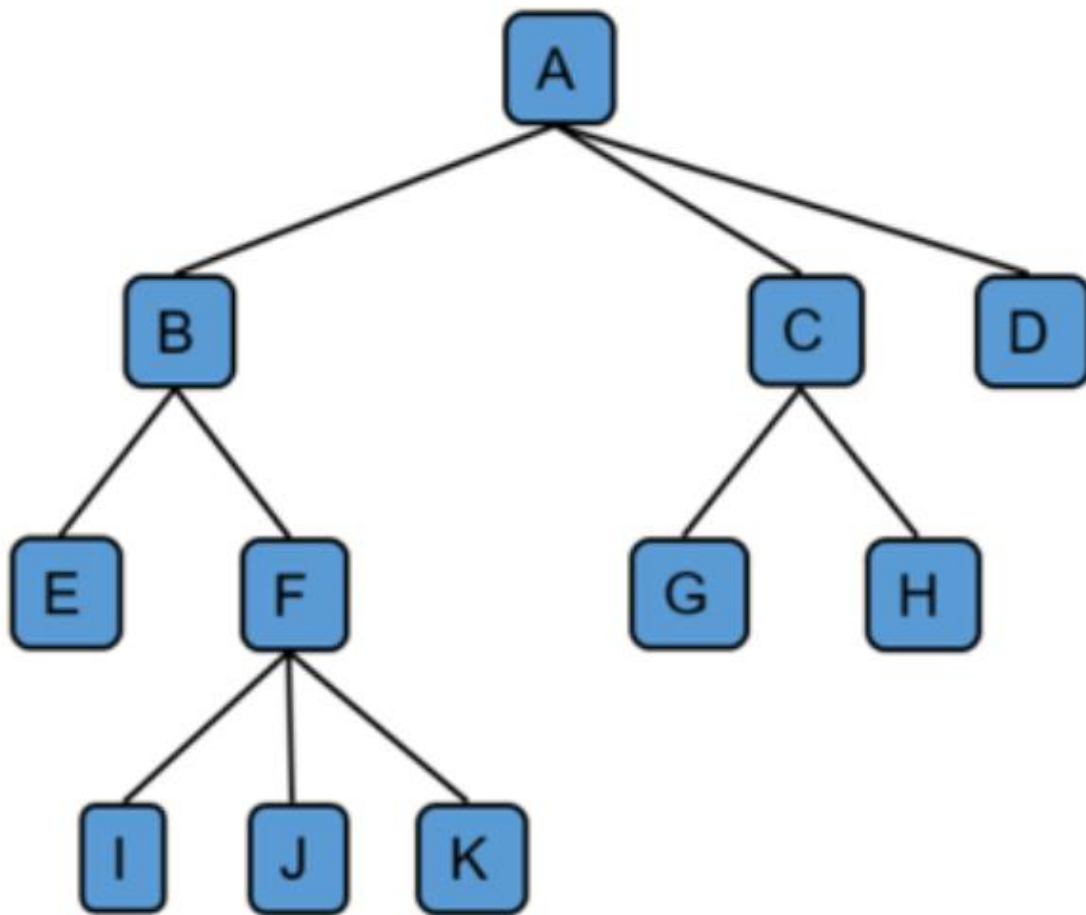
Ans:

Inorder: 4, 6, 3, 7, 2, 5, 1

Preorder: 7, 6, 4, 3, 5, 2, 1

Postorder: 4, 3, 6, 2, 1, 5, 7

Question No 10



Ans:

Preorder: A, B, E, F, I, J, K, C, G, H, D

Postorder: E, I, J, K, F, B, G, H, C, D, A